

6 Tests of Hypothesis

6.1 Introduction and Meaning of Hypothesis Testing

Motivation. In statistical inference, our goal is to make decisions or draw conclusions about a *population parameter* (such as the mean, proportion, or variance) based on information obtained from a sample. For instance, an insurance company may want to verify whether the average claim cost has increased, or an economist may test whether the average inflation rate differs from a target value.

Because we cannot usually observe the entire population, we rely on sample data to make probabilistic statements about population characteristics. **Hypothesis testing** provides a structured decision-making framework that allows us to evaluate claims or assumptions about population parameters using statistical evidence.

Definition. A **statistical hypothesis** is a formal statement or claim about one or more population parameters.

Typically, we set up two complementary hypotheses:

- **Null hypothesis** (H_0): Represents the existing belief, default position, or “no effect” situation. It is the hypothesis we assume to be true until evidence suggests otherwise.

H_0 : There is no difference or effect.

- **Alternative hypothesis** (H_1 or H_a): Represents a new belief, research claim, or suspected deviation from the null hypothesis.

H_1 : There is a significant difference or effect.

The goal of hypothesis testing is to use sample data to decide whether there is enough evidence to reject H_0 in favor of H_1 .

Illustrative Example. An insurance company advertises that the average claim amount is \$5,000. A financial analyst suspects that the true mean is different from this advertised value.

We can express this situation as:

$$H_0 : \mu = 5000, \quad H_1 : \mu \neq 5000.$$

Here:

- H_0 assumes the company’s claim is accurate.
- H_1 represents the analyst’s suspicion that the mean claim is not \$5,000.

We then collect a sample of claim data and compute a test statistic (such as a Z or t value) to determine whether the observed sample mean provides sufficient evidence against H_0 .

Logic of Hypothesis Testing. Hypothesis testing follows a systematic reasoning process, as shown below:

1. **State the hypotheses:** Formulate H_0 and H_1 based on the problem.
2. **Select the significance level (α):** Decide the maximum acceptable probability of wrongly rejecting H_0 (usually 0.05 or 5%).
3. **Compute the test statistic:** Use the sample data to calculate a standardized value (such as Z , t , or χ^2) that measures how far the sample result is from what is expected under H_0 .
4. **Find the rejection region or p -value:** Determine how extreme the test statistic must be to reject H_0 .
5. **Make a decision:**
 - If the test statistic falls into the rejection region (or $p < \alpha$), we reject H_0 .
 - Otherwise, we fail to reject H_0 .
6. **Interpret the result in context:** Conclude what the decision means in the real-life problem setting.

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Key Concepts and Terminology.

- **Significance level (α):** This represents the probability of making a **Type I error**, that is, rejecting the null hypothesis when it is actually true. Common values are $\alpha = 0.01$, 0.05 , or 0.10 .

$$P(\text{Type I error}) = \alpha.$$

For example, at $\alpha = 0.05$, we are willing to take a 5% risk of concluding that the mean claim differs from \$5,000 when it actually does not.

- **Test structure** - The test structure is generally:

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \begin{cases} \mu \neq \mu_0, & (\text{two-tailed test}) \\ \mu > \mu_0, & (\text{right-tailed test}) \\ \mu < \mu_0, & (\text{left-tailed test}) \end{cases}$$

The choice of one- or two-tailed alternative depends on the research question.

- **Type II error (β):** Occurs when we fail to reject H_0 even though it is false. The complement $1 - \beta$ is called the **power of the test**, representing the probability of correctly rejecting a false null hypothesis.

- **p -value:** The p -value is the probability, assuming H_0 is true, of obtaining a test statistic at least as extreme as the one actually observed.

$$\text{Decision rule: } \begin{cases} \text{Reject } H_0, & \text{if } p < \alpha, \\ \text{Fail to reject } H_0, & \text{if } p \geq \alpha. \end{cases}$$

- **Critical region and test statistic:** For many common tests, we compute a standardized test statistic such as Z or t .

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}, \quad t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}.$$

If the absolute value of this statistic exceeds the critical value from the standard normal (Z) or t distribution table, we reject H_0 .

- **Decision rule:**

$$\begin{cases} |Z| > Z_{\alpha/2} & \text{or} & |t| > t_{\alpha/2, \text{df}} & \Rightarrow \text{Reject } H_0, \\ |Z| \leq Z_{\alpha/2} & \text{or} & |t| \leq t_{\alpha/2, \text{df}} & \Rightarrow \text{Fail to reject } H_0. \end{cases}$$

Graphical Interpretation. When H_0 is true, the test statistic (e.g., Z) follows a known distribution (e.g., standard normal). The **rejection region** corresponds to extreme values in the tails of the distribution.

For a two-tailed test at $\alpha = 0.05$:

$$\text{Reject } H_0 \text{ if } |Z| > 1.96.$$

Graphically, the area beyond ± 1.96 (each tail having 0.025 probability) represents the 5% chance of a Type I error.

6.2 Type II Error and Power of a Statistical Test

In hypothesis testing, we evaluate evidence in data against a null hypothesis H_0 by comparing a test statistic to a critical region determined by a significance level α . However, two types of errors may occur in this decision-making process.

- **Type I error (α):** Rejecting H_0 when it is actually true.
- **Type II error (β):** Failing to reject H_0 when it is actually false.

The complement of the Type II error probability, $1 - \beta$, is known as the **power of the test**. It measures the probability that the test will correctly reject a false null hypothesis.

$$\text{Power} = P(\text{Reject } H_0 \mid H_1 \text{ is true}) = 1 - \beta$$

A powerful test is one with a high probability of detecting departures from H_0 whenever they truly exist.

6.3 Type II Error (β)

The **Type II error** occurs when we *fail to reject* the null hypothesis H_0 even though the alternative hypothesis H_1 is true. It measures the probability of not detecting a true difference.

Consider testing the hypotheses:

$$H_0 : \mu = \mu_0 \quad \text{against} \quad H_1 : \mu = \mu_1, \quad \text{where } \mu_1 > \mu_0.$$

We assume that the population variance σ^2 is known and that the sample

$$X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$$

is a random sample from a normal population.

Step 1: The Test Statistic

The test statistic for a one-sample z -test is:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

Under the null hypothesis H_0 , we have:

$$Z \sim N(0, 1),$$

since

$$E[\bar{X}] = \mu_0 \quad \text{and} \quad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Step 2: Rejection Region

For a right-tailed test at significance level α :

$$\text{Reject } H_0 \quad \text{if} \quad Z > z_{1-\alpha},$$

where $z_{1-\alpha}$ is the $(1 - \alpha)$ quantile of the standard normal distribution.

This critical value defines the rejection region:

$$\text{Rejection region: } \{Z > z_{1-\alpha}\}, \quad \text{Acceptance region: } \{Z \leq z_{1-\alpha}\}.$$

Step 3: Distribution of Z Under the Alternative Hypothesis

Under the alternative hypothesis $H_1 : \mu = \mu_1$, the sampling distribution of $\bar{X}$ is:

$$\bar{X} \sim N\left(\mu_1, \frac{\sigma^2}{n}\right).$$

Hence, the standardized test statistic becomes:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{(\bar{X} - \mu_1) + (\mu_1 - \mu_0)}{\sigma/\sqrt{n}} = \underbrace{\frac{\bar{X} - \mu_1}{\sigma/\sqrt{n}}}_{\sim N(0,1)} + \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}.$$

Therefore,

$$Z \sim N\left(\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}, 1\right).$$

This means that when H_1 is true, the mean of the Z -distribution shifts to the right by the quantity:

$$\delta = \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}},$$

which is called the **non-centrality parameter** or **standardized effect size**.

Step 4: Definition of Type II Error

A Type II error occurs when we fail to reject H_0 even though H_1 is true, that is, when Z falls inside the acceptance region while $\mu = \mu_1$. Formally:

$$\beta = P(\text{Fail to reject } H_0 \mid \mu = \mu_1) = P(Z \leq z_{1-\alpha} \mid \mu = \mu_1).$$

Step 5: Computing β Using the Alternative Distribution

Since under H_1 , $Z \sim N\left(\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}, 1\right)$, we can standardize this distribution to compute probabilities under the standard normal:

$$\begin{aligned} \beta &= P(Z \leq z_{1-\alpha} \mid \mu = \mu_1) \\ &= P\left(\frac{Z - (\mu_1 - \mu_0)/(\sigma/\sqrt{n})}{1} \leq z_{1-\alpha} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right) \\ &= \Phi\left(z_{1-\alpha} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right), \end{aligned}$$

where $\Phi(\cdot)$ denotes the cumulative distribution function (CDF) of the standard normal distribution.

Step 6: Interpretation of Components

- $z_{1-\alpha}$ — the critical cutoff chosen based on the desired significance level α .
- $\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}$ — the standardized distance between the true mean and the null mean, often denoted as the **effect size**.
- $\Phi(\cdot)$ — converts this standardized difference into a probability under the standard normal curve.

Step 7: Behavioral Insight

The formula:

$$\beta = \Phi\left(z_{1-\alpha} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right)$$

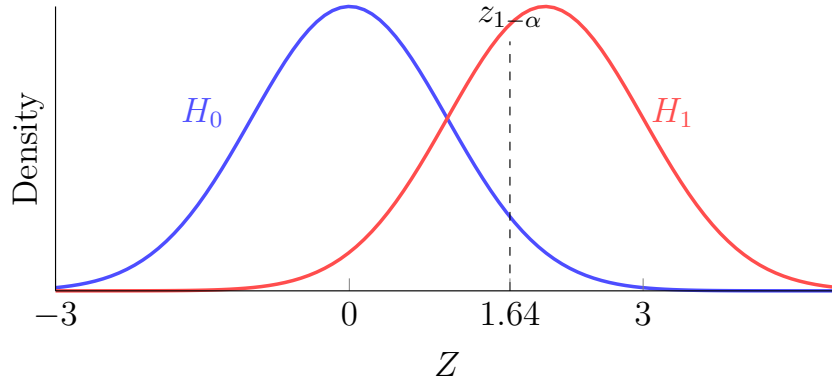
shows how β depends on the experimental design parameters:

1. Increasing n (larger sample size) decreases $\sigma/\sqrt{n}$, shifting the mean of Z further right, thereby *reducing* β .

2. Increasing the true difference $(\mu_1 - \mu_0)$ also shifts the mean rightward, lowering β .
3. Decreasing α increases $z_{1-\alpha}$, moving the critical point rightward and thereby *increasing* β .

Step 8: Graphical Illustration

Under H_0 , Z follows $N(0, 1)$, while under H_1 , Z follows $N(\delta, 1)$ with $\delta = (\mu_1 - \mu_0)/(\sigma/\sqrt{n})$.



The blue curve (H_0) shows the null distribution, while the red curve (H_1) represents the alternative. The area to the right of $z_{1-\alpha}$ under H_0 is α (Type I error), and the area to the left of $z_{1-\alpha}$ under H_1 is β (Type II error).

Step 9: Summary Expression

Thus, for a one-tailed z -test of the mean with known variance:

$$\beta = \Phi \left(z_{1-\alpha} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} \right) \quad \text{and} \quad \text{Power} = 1 - \beta.$$

This relationship forms the cornerstone for *power analysis* and *sample size determination*.

Interpretation

- The term $\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}$ is the standardized effect size.
- Larger differences between μ_1 and μ_0 increase the power.
- Increasing sample size n reduces $\sigma/\sqrt{n}$, also increasing the power.
- Decreasing α (stricter test) increases the critical value $z_{1-\alpha}$, reducing power.

Example 6.1 (Computation of Type II Error and Power). *An insurance company suspects that the average time to settle a claim has increased beyond 100 days. Historically, the mean settlement time is known to be $\mu_0 = 100$ days with a population standard deviation of $\sigma = 10$ days.*

A random sample of $n = 25$ recent claims is taken. We wish to test whether the true mean time μ has increased to $\mu_1 = 105$ days at the $\alpha = 0.05$ significance level. Compute the following:

- (a) The critical value $z_{1-\alpha}$ for a right-tailed test.
- (b) The standardized effect size $\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}$.
- (c) The probability of Type II error, β .
- (d) The power of the test, $1 - \beta$.
- (e) Interpret the results in practical terms.

Solution.

- (a) **Determine the critical value.**

For a right-tailed test at $\alpha = 0.05$,

$$z_{1-\alpha} = z_{0.95} = 1.645.$$

This value defines the cutoff point beyond which we reject H_0 in favor of H_1 .

Reject H_0 if $Z > 1.645$.

- (b) **Compute the standardized effect size.**

The standardized effect (also called the noncentrality parameter) measures how far the true mean under H_1 lies from the null mean in units of the standard error:

$$\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} = \frac{105 - 100}{10/\sqrt{25}} = \frac{5}{2} = 2.5.$$

- (c) **Compute the probability of Type II error (β).**

Recall that under H_1 , the test statistic follows:

$$Z \sim N\left(\frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}, 1\right) = N(2.5, 1).$$

The Type II error occurs when we fail to reject H_0 even though H_1 is true, i.e., when $Z \leq z_{1-\alpha}$ under this shifted distribution:

$$\beta = P(Z \leq z_{1-\alpha} \mid \mu = \mu_1) = \Phi\left(z_{1-\alpha} - \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}}\right).$$

Substituting the values:

$$\beta = \Phi(1.645 - 2.5) = \Phi(-0.855).$$

From standard normal tables (or using software):

$$\Phi(-0.855) = 0.196.$$

Hence, the probability of committing a Type II error is approximately 19.6%.

(d) **Compute the power of the test.**

The power of the test is the probability of correctly rejecting H_0 when H_1 is true:

$$1 - \beta = 1 - 0.196 = 0.804.$$

Thus, the test has approximately 80.4% power to detect an increase in mean settlement time from 100 to 105 days.

(e) **Interpretation.**

The results can be interpreted as follows:

- The test is conducted at a 5% significance level, meaning there is a 5% chance of incorrectly concluding that the mean settlement time has increased when it actually has not (Type I error).
- The Type II error probability $\beta = 0.196$ indicates there is about a 19.6% chance of failing to detect a real increase to $\mu = 105$.
- The test's power ($1 - \beta = 0.804$) means that, if the true mean has indeed increased to 105 days, there is an 80.4% chance that this test will correctly reject H_0 and detect the change.
- A power of around 0.8 is generally considered adequate in statistical practice; however, if a higher power (e.g., 0.9) were desired, one could increase the sample size n to reduce β .

Summary of Results:

$$\begin{aligned} z_{1-\alpha} &= 1.645, \\ \frac{\mu_1 - \mu_0}{\sigma/\sqrt{n}} &= 2.5, \\ \beta &= 0.196, \\ 1 - \beta &= 0.804. \end{aligned}$$

Hence, the test has about an **80.4% probability of detecting** a true increase in the mean settlement time to 105 days.

6.4 One-Sample Tests for the Mean

When we perform a hypothesis test for the population mean, our goal is to determine whether the sample evidence supports or contradicts a claim about the true average value of a variable in the population.

For example, we might test whether the average number of insurance claims per policyholder, average monthly deposits, or expected time to maturity of an investment differ from a specified benchmark.

Case 1: Population Variance Known (Z-test). This case applies when the population variance σ^2 (or equivalently the population standard deviation σ) is known — typically when the population is very large, or previous studies have established σ precisely.

Assumptions:

- (i) The sample is randomly drawn from the population.
- (ii) The population is normally distributed or the sample size n is sufficiently large (Central Limit Theorem).
- (iii) The population standard deviation σ is known.

Test statistic:

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}},$$

where:

- $\bar{X}$ = sample mean,
- μ_0 = hypothesized population mean under H_0 ,
- σ = known population standard deviation,
- n = sample size.

Under H_0 , we have:

$$Z \sim N(0, 1).$$

Decision rule (two-tailed test):

$$\text{Reject } H_0 \text{ if } |Z| > Z_{\alpha/2}.$$

For one-tailed tests:

$$\begin{cases} \text{Reject } H_0 \text{ if } Z > Z_{1-\alpha}, & \text{(right-tailed)} \\ \text{Reject } H_0 \text{ if } Z < -Z_{1-\alpha}, & \text{(left-tailed)} \end{cases}$$

p-value: The p-value is computed as:

$$\text{p-value} = \begin{cases} 2P(Z > |z_{\text{obs}}|), & \text{two-tailed} \\ P(Z > z_{\text{obs}}), & \text{right-tailed} \\ P(Z < z_{\text{obs}}), & \text{left-tailed} \end{cases}$$

If p-value $< \alpha$, we reject H_0 .

Example 6.2 ((Z-test): Testing the Average Number of Claims.). *Problem: An actuary believes the average number of insurance claims per policyholder per year is 12. From a random sample of $n = 36$ policyholders, the sample mean is $\bar{X} = 13.2$, and the population standard deviation is known to be $\sigma = 4$. At the 5% significance level, test whether the true mean number of claims differs from 12.*

Solution:

Step 1: Formulate hypotheses.

$$H_0 : \mu = 12, \quad H_1 : \mu \neq 12.$$

Step 2: Compute the test statistic.

$$Z = \frac{13.2 - 12}{4/\sqrt{36}} = \frac{1.2}{0.6667} = 1.80.$$

Step 3: Determine the critical value. For a two-tailed test at $\alpha = 0.05$, we have $Z_{0.025} = 1.96$.

Step 4: Decision.

$$|Z| = 1.80 < 1.96 \implies \text{Fail to reject } H_0.$$

Step 5: Interpretation. There is insufficient statistical evidence to conclude that the average number of claims differs from 12 at the 5% significance level. The observed difference may be due to random variation in sampling.

Case 2: Population Variance Unknown (t-test). When the population variance σ^2 is not known — which is most common in practice — we use the sample variance s^2 to estimate it and employ the *Student's t distribution* to account for additional uncertainty.

Assumptions:

- (i) The sample is a simple random sample from the population.
- (ii) The underlying population is normally distributed.
- (iii) The population variance σ^2 is unknown, but estimated by s^2 .

Test statistic:

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}.$$

Under H_0 ,

$$t \sim t_{n-1},$$

where $(n - 1)$ is the degrees of freedom (df).

Decision rule:

$$\text{Reject } H_0 \text{ if } |t| > t_{\alpha/2, n-1}.$$

For one-tailed tests:

$$\begin{cases} \text{Reject } H_0 \text{ if } t > t_{1-\alpha, n-1}, & \text{(right-tailed)} \\ \text{Reject } H_0 \text{ if } t < -t_{1-\alpha, n-1}, & \text{(left-tailed)} \end{cases}$$

p-value:

$$\text{p-value} = \begin{cases} 2P(t > |t_{\text{obs}}|), & \text{two-tailed} \\ P(t > t_{\text{obs}}), & \text{right-tailed} \\ P(t < t_{\text{obs}}), & \text{left-tailed} \end{cases}$$

Example 6.3 ((t-test): Testing Mean Monthly Deposits.). *A savings account claims that its average monthly deposit is \$1,000. A random sample of $n = 10$ customers shows:*

$$\bar{X} = 920, \quad s = 120.$$

At the $\alpha = 0.05$ level, test whether the true mean differs from \$1,000.

Solution:

Step 1: State hypotheses.

$$H_0 : \mu = 1000, \quad H_1 : \mu \neq 1000.$$

Step 2: Compute test statistic.

$$t = \frac{920 - 1000}{120/\sqrt{10}} = \frac{-80}{37.95} = -2.11.$$

Step 3: Determine critical value. For a two-tailed test with $n - 1 = 9$ degrees of freedom:

$$t_{0.025,9} = 2.262.$$

Step 4: Decision.

$$|t| = 2.11 < 2.262 \implies \text{Fail to reject } H_0.$$

Step 5: Interpretation. At the 5% level, there is no sufficient evidence to conclude that the true mean monthly deposit differs from \$1,000. The observed deviation of \$80 could be due to sampling fluctuation.

Comments and Interpretation.

- The Z -test is appropriate when σ is known or n is large ($n \geq 30$).
- The t -test is used when σ is unknown and n is small.
- As n increases, the t -distribution approaches the standard normal distribution.
- Both tests are robust to mild deviations from normality when the sample size is moderate or large.

Connection to Confidence Intervals. Hypothesis testing and confidence intervals are two sides of the same coin. For a two-sided test at significance level α , the $(1 - \alpha)$ confidence interval for μ is:

$$\bar{X} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad \bar{X} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}}.$$

If μ_0 lies outside this interval, H_0 is rejected at level α .

Practical Applications.

- In **insurance**: testing whether average claim size or frequency has changed after a new policy adjustment.
- In **banking**: testing whether the average deposit or withdrawal rate meets regulatory thresholds.
- In **financial economics**: testing whether the mean portfolio return differs from a benchmark or risk-free rate.

Summary Table: One-Sample Tests for Mean

Scenario	Test Statistic	Distribution Under H_0	Decision Rule
σ known	$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$N(0, 1)$	$ Z > Z_{\alpha/2}$
σ unknown	$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$	t_{n-1}	$ t > t_{\alpha/2, n-1}$

6.5 Two-Sample Tests for the Mean

In many real-world applications, we often need to compare the means of two populations. For example, an actuary may want to test whether two insurance products have the same average claim size, or a financial analyst may wish to compare the mean returns from two investment portfolios.

Formally, let the two populations have means μ_1 and μ_2 , and we want to test:

$$H_0 : \mu_1 = \mu_2 \quad \text{versus} \quad H_1 : \mu_1 \neq \mu_2.$$

We denote the difference in population means as $\delta = \mu_1 - \mu_2$. Under H_0 , we have $\delta = 0$.

There are three main cases depending on the assumptions about population variances and sample dependencies.

Case 1: Independent Samples, Equal Variances (Pooled t-Test). Assume that:

$$X_1, \dots, X_{n_1} \sim N(\mu_1, \sigma^2), \quad Y_1, \dots, Y_{n_2} \sim N(\mu_2, \sigma^2),$$

where both populations have the same variance σ^2 but possibly different means.

Test hypotheses:

$$H_0 : \mu_1 = \mu_2 \quad \text{vs.} \quad H_1 : \mu_1 \neq \mu_2.$$

Test statistic: Since the variances are assumed equal, we estimate σ^2 by the *pooled sample variance*:

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

This is a weighted average of the sample variances.

Then the test statistic is:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad t \sim t_{n_1+n_2-2} \text{ under } H_0.$$

Decision rule: Reject H_0 if

$$|t| > t_{\alpha/2, n_1+n_2-2}.$$

Confidence Interval: A $(1 - \alpha)$ confidence interval for $(\mu_1 - \mu_2)$ is:

$$(\bar{X}_1 - \bar{X}_2) \pm t_{\alpha/2, n_1+n_2-2} s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}.$$

If this interval does not include 0, we reject H_0 at significance level α .

Example: Comparing Two Investment Portfolios

A financial analyst wishes to determine whether two investment portfolios (A and B) differ in their average annual return.

	Portfolio A	Portfolio B
Sample size (n)	12	10
Sample mean ($\bar{X}$)	8.4	7.1
Sample variance (s^2)	1.2	1.0

We test $H_0 : \mu_1 = \mu_2$ at $\alpha = 0.05$.

$$s_p^2 = \frac{(11)(1.2) + (9)(1.0)}{20} = 1.11, \quad s_p = 1.05.$$

Compute the test statistic:

$$t = \frac{8.4 - 7.1}{1.05 \sqrt{\frac{1}{12} + \frac{1}{10}}} = \frac{1.3}{0.45} = 2.89.$$

Degrees of freedom: $df = n_1 + n_2 - 2 = 20$. From the t -table, $t_{0.025, 20} = 2.086$.

Decision: Since $|t| = 2.89 > 2.086$, we **reject** H_0 . There is significant evidence that the average return of Portfolio A is higher than that of Portfolio B.

Interpretation: The observed difference (1.3%) is statistically significant at the 5% level, suggesting that Portfolio A outperforms Portfolio B.

Case 2: Independent Samples, Unequal Variances (Welch's t-Test). When the population variances are not assumed equal, we use Welch's test, which does not pool variances.

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}},$$

and the degrees of freedom are approximated by the **Welch-Satterthwaite equation**:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{(s_1^2/n_1)^2}{n_1-1} + \frac{(s_2^2/n_2)^2}{n_2-1}}.$$

This value of df is typically non-integer, but we can round it to the nearest whole number for table lookup.

Decision rule: Reject H_0 if $|t| > t_{\alpha/2, df}$.

Remarks: Welch's test is more robust when sample sizes or variances are unequal, and is the default in most statistical software (e.g., R, Python, SPSS).

Example (Unequal Variances):

Suppose an actuary compares claim amounts from two policy types:

	Type A	Type B
n	16	9
$\bar{X}$	15.3	13.8
s^2	4.0	9.0

We test $H_0 : \mu_1 = \mu_2$ at $\alpha = 0.05$.

$$t = \frac{15.3 - 13.8}{\sqrt{4/16 + 9/9}} = \frac{1.5}{\sqrt{0.25 + 1.0}} = \frac{1.5}{1.118} = 1.34.$$

Degrees of freedom:

$$df = \frac{(1.25)^2}{(0.25^2/15) + (1^2/8)} \approx 8.4.$$

From $t_{0.025, 8} = 2.306$, since $|t| = 1.34 < 2.306$, we **fail to reject** H_0 . There is no sufficient evidence that the average claim amounts differ between the two policy types.

Case 3: Paired Samples (Dependent Observations). In some studies, the two samples are dependent or paired — for example, before-and-after studies.

Let $D_i = X_i - Y_i$ denote the paired differences. We then perform a one-sample t -test on the differences:

$$t = \frac{\bar{D}}{s_D/\sqrt{n}}, \quad t \sim t_{n-1} \text{ under } H_0 : \mu_D = 0.$$

This case is covered in detail in the section on *Paired-Sample Tests*.

Statistical Power for Two-Sample Tests. The **power** of a two-sample test measures the probability of correctly rejecting H_0 when the true mean difference $\mu_1 - \mu_2 = \delta_1 \neq 0$.

Assuming known equal variances:

$$\text{Power} = 1 - \beta = 1 - \Phi \left(t_{1-\alpha/2, df} - \frac{|\delta_1|}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \right).$$

Larger sample sizes, smaller σ^2 , and larger mean differences all increase the test's power.

Summary of Key Points:

- The pooled t -test assumes equal variances; Welch's test does not.
- Always check equality of variances using an F -test before pooling.

- If the confidence interval for $(\mu_1 - \mu_2)$ excludes zero, the difference is statistically significant.
- In finance and actuarial practice, two-sample tests are vital for comparing returns, risks, or claims between products, clients, or time periods.

6.6 Two-Sample Tests for the Mean

Case 1: Independent samples, equal variances (pooled t-test). We have two independent samples:

$$X_1, \dots, X_{n_1} \sim N(\mu_1, \sigma^2), \quad Y_1, \dots, Y_{n_2} \sim N(\mu_2, \sigma^2),$$

and want to test:

$$H_0 : \mu_1 = \mu_2 \quad \text{vs.} \quad H_1 : \mu_1 \neq \mu_2.$$

The **pooled variance** is

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}.$$

The test statistic is:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}, \quad t \sim t_{n_1 + n_2 - 2}.$$

Example: Suppose two investment portfolios are compared for their average annual return.

	Portfolio A	Portfolio B
Sample size (n)	12	10
Sample mean ($\bar{X}$)	8.4	7.1
Sample variance (s^2)	1.2	1.0

We test $H_0 : \mu_1 = \mu_2$ at $\alpha = 0.05$.

$$s_p^2 = \frac{(11)(1.2) + (9)(1.0)}{20} = 1.11, \quad s_p = 1.05.$$

$$t = \frac{8.4 - 7.1}{1.05 \sqrt{\frac{1}{12} + \frac{1}{10}}} = \frac{1.3}{0.45} = 2.89.$$

With $df = 20$, $t_{0.025, 20} = 2.086$. Since $|t| > 2.086$, we **reject** H_0 : Portfolio A has a significantly higher mean return.

Case 2: Independent samples, unequal variances (Welch's t-test). If the population variances are not assumed equal:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}},$$

with approximate degrees of freedom given by the **Welch–Satterthwaite equation**:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_2^2/n_2)^2}{n_2 - 1}}.$$

6.7 Tests for Population Proportions

One-sample proportion test. We wish to test whether the population proportion p equals some hypothesized value p_0 .

$$H_0 : p = p_0, \quad H_1 : p \neq p_0.$$

Test statistic:

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}},$$

where $\hat{p} = x/n$ is the sample proportion.

Example: A financial firm claims that 60% of customers are satisfied. A survey of 100 customers finds 52 satisfied customers. Is there evidence that the satisfaction rate differs from 60%?

$$\begin{aligned} \hat{p} &= 0.52, \quad p_0 = 0.60, \quad n = 100. \\ Z &= \frac{0.52 - 0.60}{\sqrt{0.6(0.4)/100}} = \frac{-0.08}{0.049} = -1.63. \end{aligned}$$

At $\alpha = 0.05$, $Z_{0.025} = 1.96$. Since $|Z| < 1.96$, we **fail to reject** H_0 . No significant difference in satisfaction rate is found.

Two-sample proportion test. Suppose two independent samples yield $\hat{p}_1$ and $\hat{p}_2$. We test:

$$H_0 : p_1 = p_2 \quad \text{vs.} \quad H_1 : p_1 \neq p_2.$$

Under H_0 , the pooled estimate of p is:

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}.$$

The test statistic is:

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}.$$

Example: Two insurance branches claim different policy acceptance rates. Branch A: $x_1 = 48$ out of $n_1 = 80$; Branch B: $x_2 = 60$ out of $n_2 = 100$. Test at $\alpha = 0.05$ whether the proportions differ.

$$\begin{aligned} \hat{p}_1 &= 0.60, \quad \hat{p}_2 = 0.60, \quad \hat{p} = 0.60. \\ Z &= \frac{0.60 - 0.60}{\sqrt{0.6(0.4)(1/80 + 1/100)}} = 0. \end{aligned}$$

No evidence of difference between the two branches.

Additional Examples of Tests for Population Proportions

Example 1: One-sample Proportion Test . A company advertises that at least 85% of its deliveries arrive on time. A random sample of $n = 120$ deliveries shows that $x = 96$ were on time. Is there evidence at $\alpha = 0.05$ that the true on-time delivery rate is less than 85%?

$$H_0 : p = 0.85, \quad H_1 : p < 0.85$$

$$\hat{p} = \frac{96}{120} = 0.80, \quad n = 120$$

$$Z = \frac{0.80 - 0.85}{\sqrt{0.85(1 - 0.85)/120}} = \frac{-0.05}{\sqrt{0.1275/120}} = \frac{-0.05}{0.033} = -1.52$$

At $\alpha = 0.05$, $Z_{0.05} = -1.645$. Since $Z = -1.52 > -1.645$, we **fail to reject** H_0 .
There is no sufficient evidence that the on-time delivery rate is below 85%.

Example 2: One-sample Proportion Test . A university claims that at least 70% of its graduates are employed within six months. A follow-up survey of 150 graduates finds 96 employed.

$$H_0 : p = 0.70, \quad H_1 : p < 0.70$$

$$\hat{p} = \frac{96}{150} = 0.64$$

$$Z = \frac{0.64 - 0.70}{\sqrt{0.7(0.3)/150}} = \frac{-0.06}{0.037} = -1.62$$

At $\alpha = 0.05$, $Z_{0.05} = -1.645$. Since $Z > -1.645$, we **fail to reject** H_0 .
No evidence that the employment rate is below 70%.

Example 3: One-sample Proportion Test. A manufacturer claims that 10% of its bulbs are defective. A quality control officer inspects 200 bulbs and finds 30 defective. Test whether the defect rate differs from 10%.

$$H_0 : p = 0.10, \quad H_1 : p \neq 0.10$$

$$\hat{p} = \frac{30}{200} = 0.15$$

$$Z = \frac{0.15 - 0.10}{\sqrt{0.10(0.90)/200}} = \frac{0.05}{0.021} = 2.38$$

At $\alpha = 0.05$, $Z_{0.025} = 1.96$. Since $|Z| > 1.96$, we **reject** H_0 .
There is significant evidence that the defect rate differs from 10%.

Example 4: Two-sample Proportion Test . A new advertisement campaign is introduced in Region A but not in Region B. Out of 200 customers in Region A, 120 made a purchase. Out of 180 customers in Region B, 90 made a purchase. Is there evidence that the new campaign increased purchases?

$$H_0 : p_1 = p_2, \quad H_1 : p_1 > p_2$$

$$\begin{aligned} \hat{p}_1 &= 0.60, \quad \hat{p}_2 = 0.50 \\ \hat{p} &= \frac{120 + 90}{200 + 180} = \frac{210}{380} = 0.5526 \\ Z &= \frac{0.60 - 0.50}{\sqrt{0.5526(1 - 0.5526)(1/200 + 1/180)}} \\ Z &= \frac{0.10}{\sqrt{0.247(0.01056)}} = \frac{0.10}{0.0509} = 1.96 \end{aligned}$$

At $\alpha = 0.05$, $Z_{0.05} = 1.645$. Since $Z = 1.96 > 1.645$, we **reject** H_0 .

There is evidence that the new campaign increased purchase rates.

Example 5: Two-sample Proportion Test . Two medical treatments are compared for recovery rates. Treatment A: $x_1 = 70$ recoveries out of $n_1 = 100$ patients. Treatment B: $x_2 = 60$ recoveries out of $n_2 = 120$ patients. Is there evidence that recovery rates differ?

$$\begin{aligned} H_0 : p_1 &= p_2, \quad H_1 : p_1 \neq p_2 \\ \hat{p}_1 &= 0.70, \quad \hat{p}_2 = 0.50, \quad \hat{p} = \frac{70 + 60}{100 + 120} = \frac{130}{220} = 0.591 \\ Z &= \frac{0.70 - 0.50}{\sqrt{0.591(1 - 0.591)(1/100 + 1/120)}} = \frac{0.20}{\sqrt{0.241(0.0183)}} = \frac{0.20}{0.066} = 3.03 \end{aligned}$$

At $\alpha = 0.05$, $Z_{0.025} = 1.96$. Since $|Z| > 1.96$, we **reject** H_0 .

There is a significant difference between the two treatments' recovery rates.

Example 6: Comparing Male vs. Female Voter Turnout. A political analyst wants to test if male and female voter turnout rates differ. Out of 400 sampled men, 240 voted; out of 500 sampled women, 325 voted.

$$\begin{aligned} H_0 : p_1 &= p_2, \quad H_1 : p_1 \neq p_2 \\ \hat{p}_1 &= \frac{240}{400} = 0.60, \quad \hat{p}_2 = \frac{325}{500} = 0.65 \\ \hat{p} &= \frac{240 + 325}{400 + 500} = \frac{565}{900} = 0.628 \\ Z &= \frac{0.60 - 0.65}{\sqrt{0.628(1 - 0.628)(1/400 + 1/500)}} = \frac{-0.05}{\sqrt{0.233(0.0045)}} = \frac{-0.05}{0.0324} = -1.54 \end{aligned}$$

At $\alpha = 0.05$, $Z_{0.025} = 1.96$. Since $|Z| < 1.96$, we **fail to reject** H_0 .

No evidence that male and female voter turnout rates differ.

Example 7: Two-sample Proportion Test . A traditional teaching method has a pass rate of 75%. A new method is tested on a small pilot class with 45 passes out of 70 students. Is there evidence that the new method has a **lower** pass rate?

$$H_0 : p_1 = 0.75, \quad H_1 : p_1 < 0.75$$

$$\hat{p}_1 = \frac{45}{70} = 0.643, \quad n = 70$$

$$Z = \frac{0.643 - 0.75}{\sqrt{0.75(0.25)/70}} = \frac{-0.107}{0.052} = -2.06$$

At $\alpha = 0.05$, $Z_{0.05} = -1.645$. Since $Z < -1.645$, we **reject** H_0 .

There is evidence that the new teaching method results in a lower pass rate.

6.8 Application of Hypothesis Testing in Real Life

- **Actuarial Science:** Testing whether the average claim size has increased after a policy change; comparing claim frequencies across different regions or time periods.
- **Financial Economics:** Comparing mean returns between two investment strategies; testing whether a stock's average return differs from the market benchmark.
- **Business and Policy:** Checking if customer satisfaction or loan approval rates meet targets; testing if a new marketing strategy significantly increases sales.

6.9 Summary Table

Type of Test	Population Parameter	Statistic	Distribution
One-sample mean (σ known)	μ	$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$	$N(0, 1)$
One-sample mean (σ unknown)	μ	$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$	t_{n-1}
Two-sample means (equal σ^2)	$\mu_1 - \mu_2$	$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{1/n_1 + 1/n_2}}$	$t_{n_1 + n_2 - 2}$
Two-sample means (unequal σ^2)	$\mu_1 - \mu_2$	Welch's t	approx. t
One-sample proportion	p	$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}$	$N(0, 1)$
Two-sample proportions	$p_1 - p_2$	$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}}$	$N(0, 1)$

6.10 Conclusion

Hypothesis testing provides a structured framework for making data-driven decisions. In actuarial and financial contexts, it enables professionals to:

- Verify model assumptions,
- Compare strategies or policy effects,

- Detect changes in risk behavior,
- Support decisions with quantifiable evidence.

The main steps are always:

Formulate H_0 and $H_1 \rightarrow$ Select test and significance level $\rightarrow$ Compute statistic $\rightarrow$ Make decision
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